

KHADIJAT ABUBAKAR

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## PROFESSIONAL SUMMARY

I am an AI and Data Science professional with expertise in Python, machine learning, SQL, and data visualization. I have developed impactful AI solutions in healthcare and agriculture, including a Heart Disease Prediction model with 98% accuracy and high recall, and an AI Crop Recommendation System achieving 99% accuracy.

Beyond technical expertise, I have experience mentoring learners in Python, collaborating on AI fellowship challenges, and delivering actionable insights through data analysis. As an AI Scholarship recipient at Webfala Digital Skills for All Initiative, I have gained both technical and leadership skills, preparing me to use AI in solving real-world problems and driving positive change.

## SKILLS & TOOLS

- Programming: Python, SQL
- Data Visualization: Power BI, Matplotlib, Seaborn
- Machine Learning: Classification, Regression, Model Evaluation
- Libraries: Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn
- Data Cleaning & Preprocessing
- Other Tools: Power BI, Excel, Streamlit

## PROJECTS & EXPERIENCE

AI Fellow – Webfala Digital Skills for All Initiative

- Completed intensive training in Python, machine learning, mathematics, and data science.
- Gained additional skills in leadership, communication, and problem-solving.
- Collaborated with peers on weekly AI challenges to design real-world solutions.
- Mentored individuals in Python programming, simplifying complex concepts

Heart Disease Risk Prediction

- Perform data cleaning, exploratory data analysis, feature scaling and feature selection using Python.
- Build and compare classification models (Logistic Regression, Random Forest, XGBOOST) using Scikit Learn
- Perform Hyperparameter Tuning using Gridsearch, to increase model performance.
- Evaluated performance using Accuracy, Precision, Recall and Confusion Metrics.
- Deployed the model with Streamlit for a user-friendly risk assessment

AI Crop Recommendation System

- Developed a machine learning model that recommends the most suitable crops based on soil properties (pH, nitrogen, phosphorus, potassium) and environmental factors (rainfall, temperature, humidity).
- Collected, cleaned, and preprocessed agricultural datasets to improve model accuracy.
- Implemented supervised learning algorithms and achieved reliable prediction performance.
- Designed the system to support both IoT-based data input (automated sensor readings) and manual entry for farmers without IoT access.

- Aimed at helping farmers improve productivity, reduce risk, and make data-driven agricultural decisions.

#### SOFT SKILLS

- Problem Solving | Data Story telling | Team Collaboration |
- Strong Communication Skills | Attention to Details
- Ability to learn unlearn and relearn.

#### EDUCATION & CERTIFICATIONS

Federal University of Technology | Bachelor in Mathematics Niger State, Minna

IBM Data Science Certification (Sponsored by Webfala Digital Skills for All Initiative)

Data Science Scholarship: Webfala Digital Skills for All Initiative